

Introduction to Analytics

Turning Data into Information. Information into Insight. Insight into Impact.

Analytics is the science of discovering patterns in data to understand what happened, why it happened, what is likely to happen, and what should be done. It empowers better decisions, smarter strategies, and stronger results.



1. WHAT REALLY IS ANALYTICS?

Data analytics is the process of collecting, transforming, and organizing data to identify patterns, draw conclusions, and make informed decisions. It involves a range of tools and techniques to turn raw data into actionable insights that can improve business performance, optimize processes, and predict future trends.

There are four main types:

Descriptive
What happened?


Diagnostic
Why did it happen?


Predictive
What is likely to happen?


Prescriptive
What should be done?


THE ANALYTICS PROCESS



COMMON ANALYTICS TECHNIQUES


Statistical Analysis


Regression Analysis


Time Series Analysis


Data Mining


Machine Learning

These techniques help transform raw data into meaningful insights and predictive power.

TYPES OF ANALYTICS (D2P3)

**Descriptive Analytics**
Summarizes past data to understand what has happened.

**Diagnostic Analytics**
Investigates the root causes of past events.

**Predictive Analytics**
Uses historical data to forecast future trends and outcomes.

**Prescriptive Analytics**
Recommends a specific course of action to achieve a desired outcome.

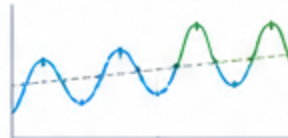
PATTERNS THAT DATA CAN REVEAL

A dataset can reveal a wide range of patterns. Identifying these patterns is a core goal of analytics.

1 Trend Analysis

Describes the general direction of a variable over time.

- Upward trend: increasing
- Downward trend: decreasing
- Horizontal trend: stable

2 Seasonal & Cyclical Patterns

Fluctuations that repeat over regular intervals.

- Seasonality:** fixed periods (day, week, month, year)
- Cyclicity:** longer, less predictable economic cycles

3 Correlation & Relationships

Statistical relationships between variables.

- Positive correlation (↑↑)
- Negative correlation (↑↓)
- No correlation

4 Clustering & Segmentation

Grouping similar data points based on characteristics.

- Market segmentation
- Behavioral patterns
- Targeted strategies

5 Anomalies & Outliers

Data points that deviate significantly from others.

- Fraud detection
- System monitoring
- Outliers and errors

6 Association & Frequent Itemsets

Items that frequently occur together.

- Market basket analysis
- Web usage mining
- Common sequences

ANALYTIC TOOLS I KNOW


Microsoft Excel & Access


Python & R


Power BI & Tableau


Data Organization & Transformation

- Structured Data:** SQL Database, Wonderware, PostgreSQL
- Unstructured Data:** MongoDB
- Big Data:** Apache Spark, Apache Hadoop, XX

WHEN IS ANALYTICS INVOLVED?



HOW IS ANALYTICS DIFFERENT?

Analysis is a subset of analytics. With analysis, you have the data in front of you, you don't really care how it got there or where it came from. Analysis is about using the tools and techniques at your disposal to get meaning and actionable insight from the dataset.

Data science is about using data to tell stories. But before they can narrate the stories, data scientists have to cleanse the data, explore it, model it, and then communicate it.

In short: Data Science is the broader discipline, Analytics is the process, and Analysis is the focused examination.



Data Science
The broad field of studying data to extract knowledge. Involves data collection, engineering, modeling, machine learning, and communication.

Data Analytics
The process of examining data to find patterns, draw conclusions, and support decision-making. Includes analysis, visualization, and reporting.

Data Mining
A specific set of techniques used within analytics to discover hidden patterns and relationships in large datasets.